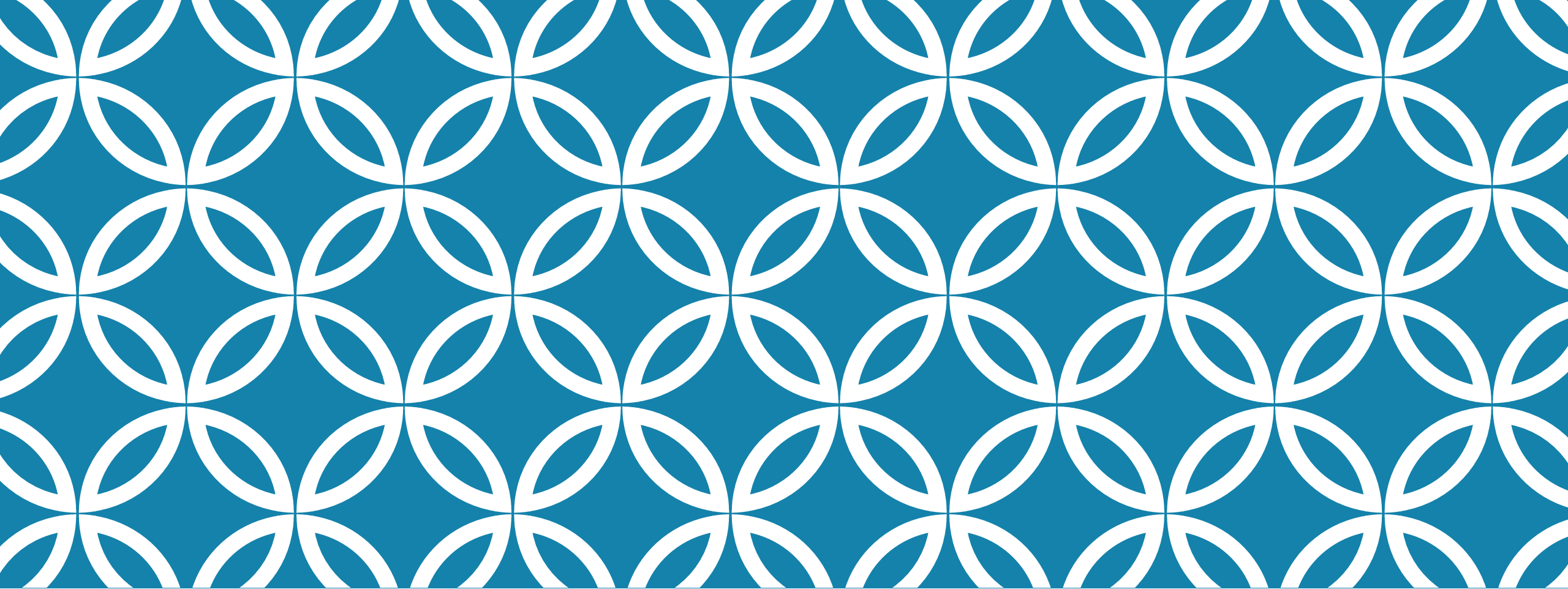




PIPELINING I

WHAT IS PIPELINING?

Pipelining is an **implementation technique** whereby multiple **instructions** are overlapped in execution

Do you know what an instruction is, and how it gets executed?

It takes advantage of parallelism that exists among the actions needed to execute an instruction.

Today, pipelining is the key implementation technique used to make **fast CPUs**.

REAL WORLD ANALOGY

A pipeline is like an assembly line.

In an automobile manufacturing line, there are many steps, each contributing something to the construction of the car.

Each step operates in parallel with the other steps, although on a different car.

REAL WORLD ANALOGY

Automobile manufacturing line without pipelining, in 3 stages

Assemble (A), Paint (P), Fix tires (T)

Assuming A+P+T take 3 hours

Time line	0-3 hours	3-6 Hours	6 -9Hours
Car 1	A+P+T		
Car 2		A+P+T	
Car 3			A+P+T

REAL WORLD ANALOGY

Automobile manufacturing line with pipelining, in 3 stages

Assemble (A), Paint (P), Fix tires (T)

Assuming each stage takes 1 hour

It takes 5 hours with pipelining and 9hours without pipelining

Speedup = $9/5$

Time line	1 st hour	2 nd hour	3 rd hour	4 th hour	5 th hour	6 th hour	7 th hour	8 th hour	9 th hour
Car 1	A	P	T						
Car 2		A	P	T					
Car 3			A	P	T				

REAL WORLD ANALOGY

Here our **assumptions** are:

1. There is no time consumed to transition car from one working station on other
2. All the stages consume equal time

However these **assumption** are **not valid**

1. Each stage can take different amount of time
 - For example, painting a car takes 2 hours and Assembly and installing types take 0.5 hours each
2. To transfer car from one stage to another will take time.
 - For example, it takes 10 minutes to transfer a car from Assembly room to paint rooms and 10 minute to transfer from paint room to tire fixing room.
 - Note that this will not be consume in un-pipeline factory as everything will be done is same room (this is the overhead of using pipelining)

REAL WORLD ANALOGY

Automobile manufacturing line with pipelining, in 3 stages

If Assemble (A), Paint (P), Fix tires (T) take different time

If Assembly take 0.5 hour, Paint 2 hours and Tires 0.5 hours

□ Paint of Car2 cannot start before 2.5th hour because Paint unit is busy with Car1. So after Assembly Car 2 has to wait for 1.5 Hours

□ Total time take to complete 3 instructions =7 hours

Speed Up = 9/7

Time line	0-0.5h	0.5-2.5h	2.5-4.5h	4.5-6.5 h	6.5-7 h
Car 1	A (0.5 h)	P (2h)	T (0.5h)		
Car 2		A (0.5 h) +wait for 1.5 h	P (2h)	T (0.5h)	
Car 3			A (0.5h)+ wait for 1.5 h	P (2h)	T (0.5h)

REAL WORLD ANALOGY

Automobile manufacturing line with pipelining, in 3 stages

If we add time for transition between stages as well

Total time taken to complete 3 instructions = 7h+40m=7.67h

Speed Up = $9/7.67$

Time line	0-0.5h	Transition time 10 min	0.5 +10m-2.5h +10m	Transition time 10 min	2.5h+20m-4.5h+2 0m	Transition time 10 min	4.5h+30 m-6.5 h+30m	Transition time 10 min	6.5h+40m -7 h+40m
Car 1	A (0.5 h)		P (2h)		T (0.5h)				
Car 2			A (0.5 h) +wait for 1.5 h		P (2h)		T (0.5h)		
Car 3					A (0.5h)+ wait for 1.5 h		P (2h)		T (0.5h)

REAL WORLD ANALOGY::THROUGHPUT AND LATENCY

In case of car manufacturing **throughput** is **number of cars create per hour**

- ❑ From slide 8, throughput **with** pipelining is 3 Cars is 7.67 Hours ~ 0.4 Cars/hour
- ❑ From slide 4, throughput **without** pipelining is 3 Cars is 9Hours ~ 0.33 Cars/hour
- ❑ So we can see that the throughput with pipelining is higher.

Latency is **time taken to complete one car**

- ❑ From slide 8, latency **with** pipelining is 4.5 hours (4.5 hours consumed to create car 2 and 3)
- ❑ From slide 4, latency **without** pipelining is 3 hours (3 hours consumed to create each car)
- ❑ Thus we can see that latency is worse with pipelining

Observation: Pipeline does not improve latency, it only improves throughput (observe the bottlenecks)

THROUGHPUT AND LATENCY FOR N=3

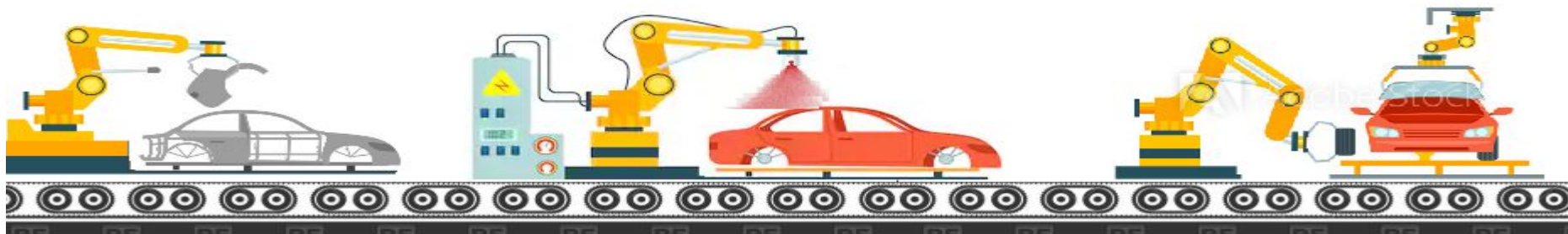
Case	Latency	Throughput	Speed up
Non pipeline	3	$3/9=0.33$	NA
Pipeline ideal case	3	$3/5=0.6$	$9/5$
Pipeline with all stages not of equal time	4.5	$3/7=0.42$	$9/7$
Non ideal case with latch time	4.833	$3/7.67=0.39$	$9/7.67$

REAL WORLD ANALOGY:: PROCESS CYCLE

The time required between moving an object one step down the pipeline is a **processor cycle**

Assume that all stages proceed at the same time

- There is only one conveyer belt and it only moves cars to next step when longest stage is complete, i.e., every two hours.
- Processor cycle will be 2 hours (See we're trying to define it – think of the computer clock – the smallest unit needs to be followed)
- Even the car in T stage will be there for 2 hours, and will only come out of pipeline when 2 hours of P (of other instruction are completed)



REAL WORLD ANALOGY:: PROCESS CYCLE

Consider process cycle to be 2 hours, throughput with pipelining is 3 Cars/ 10 Hours
~0.3 Cars/hour

□ This is even worse than throughput as compared with an un-pipelined version

This issue can be resolve if stages are more or less of equal time duration (paint job can be further divided into smaller stages)

Time line	2 hours	2 hours	2 hours	2 hours	2 hours
Car 1	A	P	T		
Car 2		A	P	T	
Car 3			A	P	T

EXERCISE

Divide your paint job in 4 stages, 1st coat, 2nd coat, 3rd coat and drying , each phase takes 0.5 hours.

What will be throughput with these 4 stages for 3 cars?

Ignore the latch/transition time